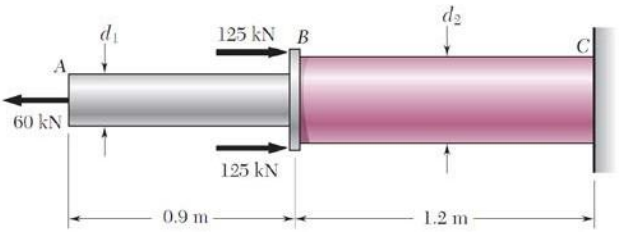
Question 1

Two solid cylindrical roads AB and BC are welded together at B and loaded as shown. Knowing that $d_1 = 30$ mm and $d_2 = 50$ mm, find the average normal stress in the mid-section of (a) rod AB, (b) rod BC.



SOLUTION

(a) rod AB

Force: $P = 60 \times 10^3$ N tension

Area: $$A = \frac{\pi}{4} d_1^2 = \frac{\pi}{4} (30 \times 10^{-3})^2 = 706.86 \times 10^{-6} \text{ m}^2$$

Normal stress: $$\sigma_{AB} = \frac{P}{A} = \frac{60 \times 10^3}{706.86 \times 10^{-6}} = 84.88 \times 10^6 \text{ Pa}$$

$$\sigma_{AB} = 84.9 \text{ MPa}$$ ◄

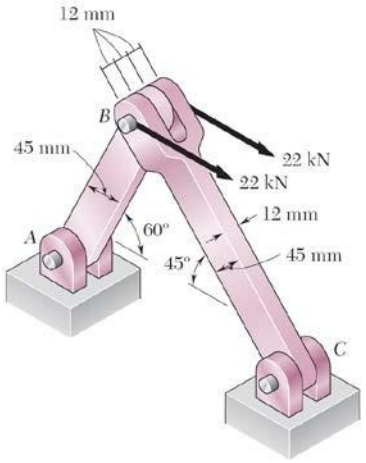
(b) rod BC

Force: $$P = 60 \times 10^3 - (2)(125 \times 10^3) = -190 \times 10^3 \text{ N}$$

Area: $$A = \frac{\pi}{4} d_2^2 = \frac{\pi}{4} (50 \times 10^{-3})^2 = 1.9635 \times 10^{-3} \text{ m}^2$$

Normal stress: $$\sigma_{BC} = \frac{P}{A} = \frac{-190 \times 10^3}{1.9635 \times 10^{-3}} = -96.77 \times 10^6 \text{ Pa}$$

Question 2

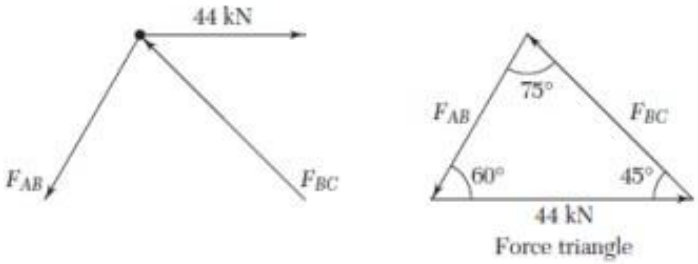


Two horizontal 22 kN forces are applied to pin B of the assembly shown. Knowing that a pin of 20 mm diameter is used at each connection, determine the maximum value of the average normal stress

(a) in link AB,
(b) in link BC.

SOLUTION

Use joint B as free body.



Force triangle

Law of Sines

$$\frac{F_{AB}}{\sin 45^\circ} = \frac{F_{BC}}{\sin 60^\circ} = \frac{44}{\sin 75^\circ}$$

$$F_{AB} = 32.2 \text{ kN} \qquad F_{BC} = 39.4 \text{ kN}$$

Link AB is a tension member

Minimum section at pin $A = (45)(12) = 540 \text{ mm}^2$

(a) Stress in AB $\sigma_{AB} = \dfrac{F_{AB}}{A_{net}} = \dfrac{32.2\,(1000)}{540} = 59.6 \text{ N/mm}^2$ ◄

or $= 59.6$ MPa

Link BC is a compression member

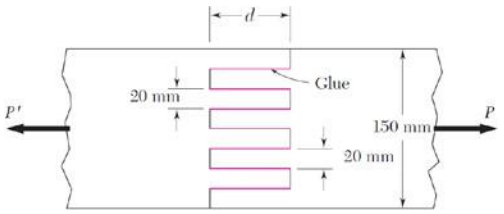
Cross sectional area is $A = (45)(12) = 540 \text{ mm}^2$

(b) Stress in BC $\sigma_{BC} = \dfrac{-F_{BC}}{A} = \dfrac{-39.4(1000)}{540} = -72.96 \text{ N/mm}^2$

or -72.95 MPa ◄

Question 3

Two wooden planks, each 22 mm thick (into the page) and 150 mm wide, are joined by the glued mortise joint shown. Knowing that the joint will fail when the average shearing stress in the glue reaches 0.8 MPa, determine the smallest allowable length d of the cuts if the joint is to withstand an axial load of magnitude P = 5.4 kN.



SOLUTION

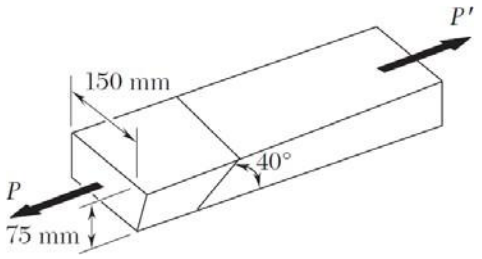
Seven surfaces carry the total load $P = 5.4$ kN

$$\text{Area } A = (7)\,(22)\,d = 154\,d$$

$$\tau = \frac{P}{A} \quad \therefore \quad A = \frac{P}{\tau} \qquad 154\,d = \frac{5400}{0.8} \qquad d = 43.8 \text{ mm}$$

Question 4

Two wooden members of 75 x 150 mm uniform rectangular cross section are joined by the simple glued scarf splice shown. Knowing that P = 10 kN, determine the normal and shearing stresses in the glued splice.



SOLUTION

$$\theta = 90° - 40° = 50° \qquad P = 10 \text{ kN}$$

$$A_o = (75)(150) = 11250 \text{ mm}^2$$

$$\sigma = \frac{P}{A_o}\cos^2\theta = \frac{(10\,000)\cos^2 50°}{11250} = 0.367 \text{ N/mm}^2$$

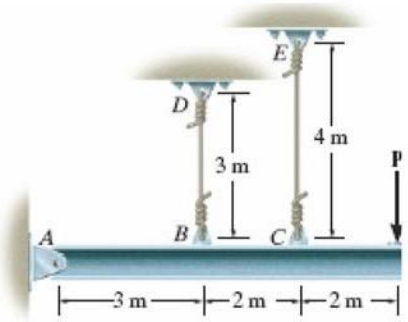
$$\sigma = 0.367 \text{ MPa}$$

$$\tau = \frac{P}{2A}\sin 2\theta = \frac{(10\,000)\sin 100°}{(2)(11250)} = 0.438 \text{ N/mm}^2$$

$$\tau = 0.438 \text{ MPa}$$

Question 5

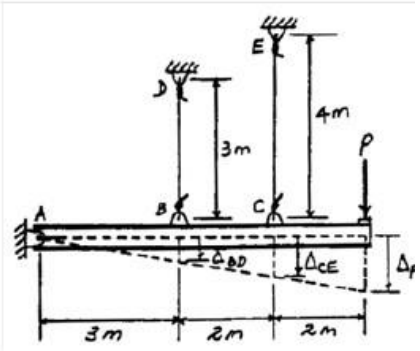
The rigid beam is supported by a pin at A and wires BD and CE. If the maximum allowable normal strain in each wire is ε_{max} = 0.002 mm/mm, determine the maximum vertical displacement of the tip of the beam.



Solution:

$$\frac{\Delta L_{BD}}{a} = \frac{\Delta L_{CE}}{a+b} \qquad \frac{\Delta L_{CE}}{a+b} = \frac{\Delta_{tip}}{a+b+c}$$

Average Elongation/Vertical Displacement:



$\Delta L_{BD} := L_{BD} \cdot \varepsilon_{allow}$ $\quad \Delta L_{BD} = 6.00\,mm$

$$\Delta_{tip} := \left(\frac{a+b+c}{a}\right) \cdot \Delta L_{BD}$$

$\Delta_{tip} = 14.00\,mm$

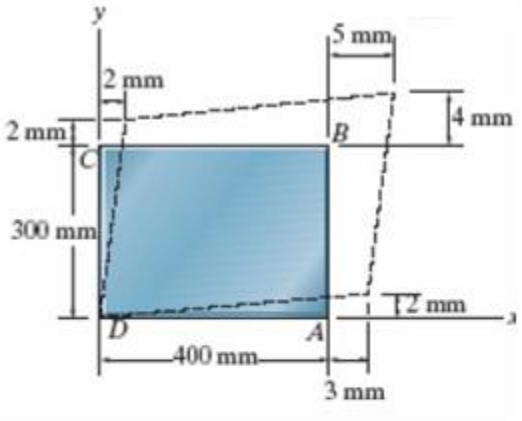
$\Delta L_{CE} := L_{CE} \cdot \varepsilon_{allow}$ $\quad \Delta L_{CE} = 8.00\,mm$

$$\Delta_{tip} := \left(\frac{a+b+c}{a+b}\right) \cdot \Delta L_{CE}$$

$\Delta_{tip} = 11.20\,mm$ **(Controls !)** **Ans**

Question 6

The piece of plastic is originally rectangular. Determine the magnitudeof the shear strain γ_{xy} at corner D if the plastic distorts as shown by thedashed lines.



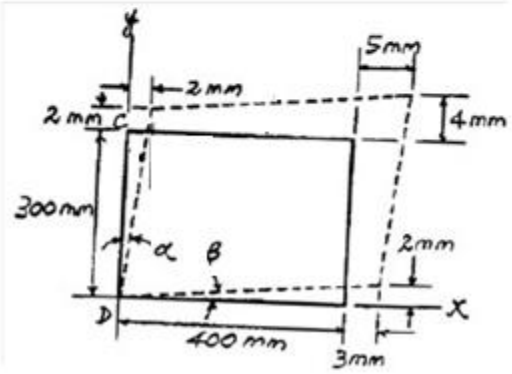
Solution

Note that sine the angles are small (in *Mechanics of Materials*, we always consider small deformation), we get:

$$\alpha = \tan\alpha = \frac{2}{300+2} = 0.00662$$

$$\beta = \tan\beta = \frac{2}{400+3} = 0.00496$$

$$\gamma_D = \alpha + \beta = 0.0116$$



Note:
Because the deformation is small, the shear strain can also be calculated as:

$$\alpha = \tan\alpha = \frac{2}{300} = 0.00667$$

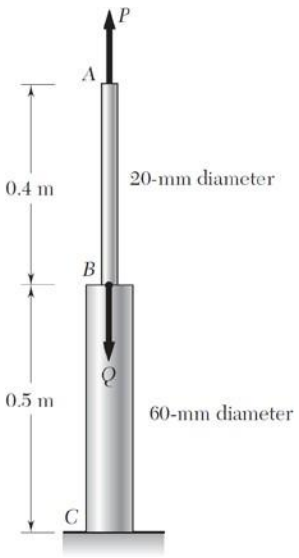
$$\beta = \tan\beta = \frac{2}{400} = 0.00500$$

$$\gamma_D = \alpha + \beta = 0.0117$$

Question 7

Both portions of the rod ABC are made of aluminium for which E = 70 GPa. Knowing that the magnitude of P is 4 kN, determine

(a) the value of Q so that the deflection at A is zero,
(b) the corresponding deflection of B.



SOLUTION

(a) $A_{AB} = \frac{\pi}{4} d_{AB}^2 = \frac{\pi}{4} (0.020)^2 = 314.16 \times 10^{-6} \text{ m}^2$

$A_{BC} = \frac{\pi}{4} d_{BC}^2 = \frac{\pi}{4} (0.060)^2 = 2.8274 \times 10^{-3} \text{ m}^2$

Force in member AB is P tension

$$\text{Elongation } \delta_{AB} = \frac{PL_{AB}}{EA_{AB}} = \frac{\left(4 \times 10^3\right)(0.4)}{\left(70 \times 10^9\right)\left(314.16 \times 10^{-6}\right)}$$

$$= 72.756 \times 10^{-6} \text{ m}$$

Force in member BC is $Q - P$ compression

$$\text{Shortening } \delta_{BC} = \frac{(Q - P)L_{BC}}{EA_{BC}} = \frac{(Q - P)(0.5)}{(70 \times 10^9)(2.8274 \times 10^{-3})}$$

$$= 2.5263 \times 10^{-9} (Q - P)$$

For zero deflection at A $\quad \delta_{BC} = \delta_{AB}$

$$2.5263 \times 10^{-9} (Q - P) = 72.756 \times 10^{-6} \quad \therefore \quad Q - P = 28.8 \times 10^3 \text{ N}$$

$$Q = 28.3 \times 10^3 + 4 \times 10^3 = 32.8 \times 10^3 \text{ N} = 32.8 \text{ kN}$$

(b) $\delta_{AB} = \delta_{BC} = \delta_B = 72.756 \times 10^{-6} \text{ m} = 0.0728 \text{ mm}$